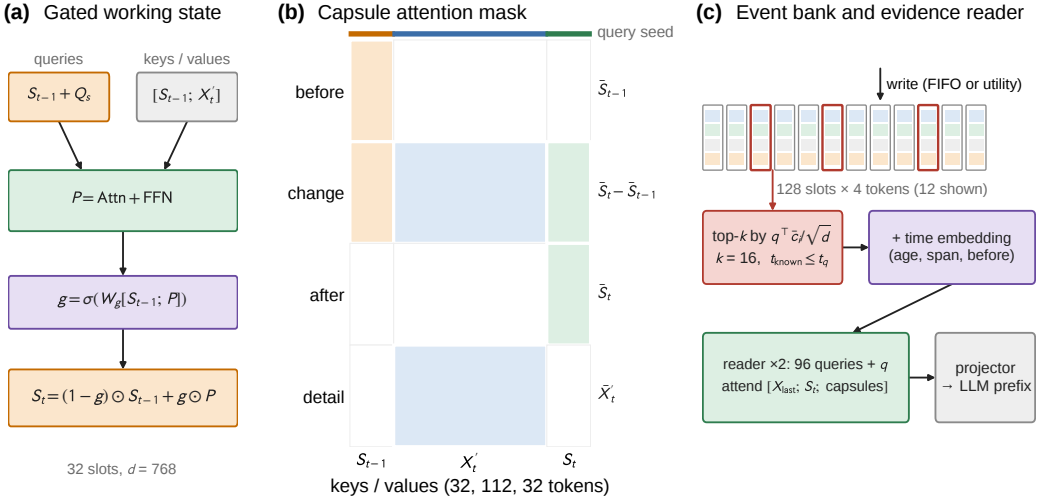
(a) Gated working state
queries
keys / values
$S_{t-1} + Q_s$
$[S_{t-1};\ X'_t]$
$P = \mathrm{Attn} + \mathrm{FFN}$
$g = \sigma(W_g[S_{t-1};\ P])$
$S_t = (1-g) \odot S_{t-1} + g \odot P$
32 slots, $d = 768$
(b) Capsule attention mask
query seed
before
change
after
detail
$\bar{S}_{t-1}$
$\bar{S}_t - \bar{S}_{t-1}$
$\bar{S}_t$
$\bar{X}'_t$
S_{t-1}
X'_t
S_t
keys / values (32, 112, 32 tokens)
(c) Event bank and evidence reader
write (FIFO or utility)
128 slots × 4 tokens (12 shown)
top-k by $q^\top \bar{c}/\sqrt{d}$
$k = 16$, $t_{known} \le t_q$
+ time embedding
(age, span, before)
reader ×2: 96 queries + q
attend $[X_{last};\ S_t;$ capsules$]$
projector
→ LLM prefix